Generate Div Tags

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The problem

- All valid strings of n matched pairs (balanced-parentheses generation).

Intuition

- Backtracking: you can open a tag while some remain and close one only if more are open than closed.

Brute force vs optimal

- Generate all 2^(2n) strings and filter → $O(4^n \cdot n)$
- Backtracking with counts → O(Catalan(n))
- Output size is the Catalan number

Approach

- 1. Track opening and closing tags left
- 2. Add an opening tag if available
- 3. Add a closing tag if it keeps balance
- 4. Record complete strings

Worked example

```
state: opens left, closes left

add <div> if opens > 0

add </div> if closes > opens
```

Two rules keep tags valid

Worked example

```
(((   →   (((()   →   (((())   →   (((()))
```

Depth-first: open, open, open, close...

Worked example

<<<>>> <<><>> <<>><> <><<>> <><><>

n = 3 → 5 combinations

Java solution

```java
public class GenerateDivTags {
    public static ArrayList<String> generateDivTags(int n) {
        ArrayList<String> out = new ArrayList<>();
        gen(n, n, new StringBuilder(), out);
        return out;
    }

    private static void gen(int open, int close, StringBuilder sb, List<String> out) {
        if (close == 0) { out.add(sb.toString()); return; }
        int len = sb.length();
        if (open > 0) { sb.append("<div>"); gen(open - 1, close, sb, out); sb.setLength(len); }
        if (close > open) { sb.append("</div>"); gen(open, close - 1, sb, out); sb.setLength(len); }
    }
}
```

Complexity

- Time: O(Catalan(n) * n)
- Space: O(Catalan(n) * n)

Edge cases & pitfalls

- Close only when closes > opens
- N = 0 → [""]
- Undo appends

Interview tips & follow-ups

- Mention Catalan numbers

Recap

- Backtracking: you can open a tag while some remain and close one only if more are open than closed.
- Complexity: O(Catalan(n) * n)